www.tntextbooks.in

Internal structure

T.S. of Root

The internal organization of the primary root reveals the following parts. 1. Epiblema, 2. Cortex 3. Vascular region (Figure 2.41). Epiblema is the outermost layer and is made up of single layered parenchyma. It is followed by thin walled parenchymatous cortex. The cortex is delimited by single layered endodermis. A multilayered parenchymatous pericycle is present and it surrounds the vascular tissue. The xylem is diarch in young root and tetrarch in older ones. Secondary growth is present. Coralloid root also shows similar structure but the middle cortex is characterized by the presence of Algal zone. Blue green alga called, *Anabaena* is found in this zone. The xylem is triarch and exarch.

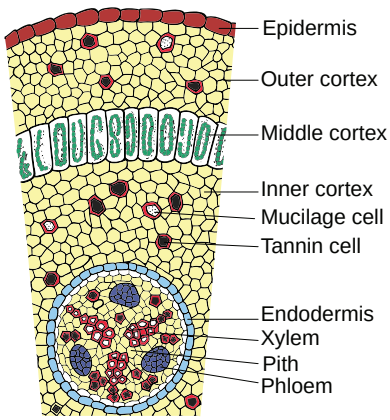


Figure 2.41: T.S. of Coralloid root

T.S. of Stem

The cross section of young stem is irregular in outline due to the presence of persistent leaf bases. It is differentiated into epidermis, cortex and vascular cylinder. It resembles the structure of a dicot stem (Figure 2.42).

The epidermis is the outermost layer and is covered with thick cuticle. It is discontinuous due to the presence of leaf bases. The cortex constitutes the major part and is made up of thin walled parenchymatous cells. The cells are filled with starch grains. Cortex also possesses several mucilage ducts and tannin cells. In young stem the vascular bundles are arranged in the form of a ring. A broad medullary ray is present. The vascular bundles are conjoint, collateral, endarch and open. Xylem is made up of tracheids and phloem consists of sieve tubes and phloem parenchyma. Companion cells are absent. The cambium present in the vascular bundle is active for short period. The secondary cambium is formed from the pericycle or cortex and helps in secondary growth of the stem. The cortical region shows a large number of leaf traces. The presence of direct leaf traces and girdling leaf trace is the unique feature of *Cycas* stem. Secondary growth results in polyxylic condition. Phellogen and cork are formed and replace the epidermis.The wood formed belongs to manoxylic type.

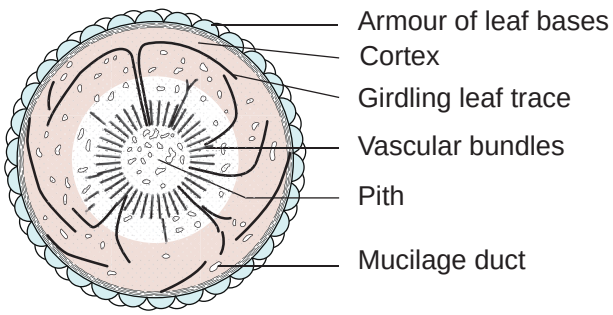


Figure 2.42: T.S. of stem

T.S. of Rachis

The outermost layer is epidermis and is covered by thick cuticle. The hypodermis is made up of two layers of sclerenchyma on the adaxial side and many layered on the abaxial side. The ground tissue is parenchymatous. The peculiar feature